Deconvolution and phylogeny inference of diverse variant types integrating bulk DNA-seq with single-cell RNA-seq

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Abstract & Overview

- **Goal:** Reconstruct tumor clonal history and mutation profiles accurately.
- **Challenge:** Single-cell sequencing has high dropout rates and limited variant coverage.
- **Solution (TUSV-INT):** A novel computational framework integrating bulk DNA-seq and single-cell RNA-seq (scRNA-seq).
- **Key Capabilities:** Covers Single Nucleotide Variants (SNVs), Copy Number Alterations (CNAs), and Structural Variants (SVs).
- **Core Engine:** Driven by an Integer Linear Programming (ILP) model.

Introduction & Challenges

Cancer Evolution

- Tumors are heterogeneous and evolve over time through accumulated mutations.
- Understanding this lineage is crucial for targeted therapy.

Limitations of Current Sequencing

- *scRNA-seq:* Sparse coverage, high dropout rates, low resolution for CNAs, and very limited SV detection.
- *Bulk DNA-seq:* Cannot directly identify co-occurring mutations in single cells.
- **Conclusion:** Multiomic integration is essential to bridge these gaps.

Proposed Solution: The TUSV-INT Approach

- **What is TUSV-INT?** An integrative tool combining the depth of bulk DNA-seq with the cellular resolution of scRNA-seq.

Inputs required:

- Variant Allele Frequencies (VAFs) from bulk DNA.
- Mutation calls (presence/absence) from scRNA-seq cells.

Expected Outputs:

- Inferred clonal tree (phylogeny).
- Assignment of mutations to specific clones.
- Estimation of clonal frequencies across samples.

Methodology: Objective Function

The Mathematical Core

$$\min_{U,C,M} \|F - UC\| + \|C^0 - MC_{RNA}\| + \lambda_1 R + \lambda_2 S$$

Key Parameters (See Table 1):

- F: Variant frequencies (Bulk DNA)
- U: Clonal frequency matrix
- C: Clone profile matrix
- C^0: Unobserved true scRNA-seq matrix
- M: Cell-to-clone assignment matrix
- λ_1, λ_2: Regularization terms for phylogenetic constraints

Parameters

Notation	Interpretation	Type
n	The number of DNA clones	User defined
n_r	The number of RNA clones	Data derived
m	The number of samples	Data derived
l, g, r	The number of SV breakpoints, SNVs, and DNA copy number segments, respectively	Data derived
$F \in \mathbb{R}_{\geq 0}^{m \times (l+g+2r)}$	$F_{i,j}$: Average copy number of variant j in sample i	Constant
$Q \in \{0,1\}^{(l+g) \times r}$	$Q_{i,j} = 1$ iff breakpoint or SNV i is in segment j	Constant
$G \in \{0,1\}^{l \times l}$	$G_{i,j} = 1$ iff i and j are paired breakpoints, deriving from the same SV	Constant
$C^{\mathrm{RNA}} \in \mathcal{Z}_{\geq 0}^{n_r \times (2r)}$	$C_{i,j}^{\mathrm{RNA}}$: Clonal copy number of segment j in RNA clone i	Constant
$C' \in \mathcal{Z}_{\geq 0}^{n \times (2r)}$	$C'_{i,j}$: Clonal copy number of segment j in clone i of the phylogeny	Variable
$U \in \mathbb{R}_{\geq 0}^{m \times n}$	$U_{i,j}$: Fraction of clone j in sample i	Variable
$C \in \mathbb{Z}_{\geq 0}^{n \times (l+g+2r)}$	$C_{i,j}$: Clonal copy number of variant j in clone i	Variable
$M \in \{0,1\}^{n \times n_r}$	$M_{i,j} = 1$ iff clone i in the phylogeny represents RNA clone j	Variable
$E \in \{0,1\}^{n \times n}$	$E_{i,j} = 1$ iff there is a directed edge from clone i to clone j	Variable
$A \in \{0,1\}^{n \times n}$	$A_{i,j} = 1$ iff clone i is an ancestor of clone j	Variable
$W \in \{0,1\}^{n \times n \times (l+g)}$	$W_{i,j,b} = 1$ iff breakpoint or SNV b is mapped on the edge from clone i to clone j	Variable
$D \in \{0,1\}^{(l+g)}$	$D_b = 1$ iff breakpoint or SNV b belongs to the first allele	Variable

Methodology: Optimization Process

- **Optimization Strategy:** Constrained optimization solved via *Coordinate Descent*.
- **Iterative Process:**
  - Fix U, C to optimize M, C^0.
  - Fix M, C^0 to optimize U, C.
- **Robustness:** Uses $L1$ norms ($\| \cdot \|_1$) to reduce the impact of extreme outliers.
- **Computational Efficiency:** Employs Integer Linear Programming (ILP). To handle large scRNA-seq datasets, *cell subsampling* is applied to maintain tractability.

Algorithm: Objective Formulation

1. Main Objective Function

$$\min_{U,C,M} |F - UC| + |C^0 - MC_{RNA}| + \lambda_1 R + \lambda_2 S \qquad (1)$$

2. Mixture Fraction Constraints

To ensure biological validity, the clonal mixture fractions for each sample i must sum to 1:

$$\sum_{j=1}^{n} U_{i,j} = 1 \qquad \forall i \in \{1, \ldots, m\} \qquad (2)$$

Optimization Strategy: The problem is solved using **Coordinate Descent**. It alternates between optimizing U and optimizing $\{C, M\}$, formulating each subproblem as an Integer Linear Program (ILP) using the Gurobi solver.

scRNA-seq Mapping Constraints (M)

The matrix M maps nr scRNA-seq clusters to n inferred bulk clones.

1. Mapping Properties:

$$M_{i,j} \in \{0,1\} \qquad \forall i \in \{1,\ldots,n\},\ j \in \{1,\ldots,nr\} \tag{3}$$

$$\sum_{j=1}^{nr} M_{i,j} = 1 \qquad \forall i \in \{1,\ldots,n\} \quad \text{(Each clone maps to 1 RNA cluster)} \tag{4}$$

$$\sum_{i=1}^{n} M_{i,j} \leq 1 \qquad \forall j \in \{1,\ldots,nr\} \quad \text{(RNA cluster maps to at most 1 clone)} \tag{5}$$

scRNA-seq Mapping Constraints (M)

2. L1 Distance Formulation for RNA Copy-Number:

To linearize the objective $|C^0 - MC_{RNA}|$ for ILP, auxiliary variables C^{Δ} are used:

$$C^{\Delta}_{i,j} \geq C^{0}_{i,j} - \sum_{k=1}^{nr} M_{i,k}\, C^{RNA}_{k,j} \tag{6}$$

$$C^{\Delta}_{i,j} \geq -C^{0}_{i,j} + \sum_{k=1}^{nr} M_{i,k}\, C^{RNA}_{k,j} \tag{7}$$

$$|C^0 - MC_{RNA}| = \sum_{i=1}^{n} \sum_{j=1}^{2r} C^{\Delta}_{i,j} \tag{8}$$

Algorithm: Phylogenetic Tree Structure

The tumor phylogeny is modeled as a rooted directed binary tree. $E_{i,j}$ represents a directed edge, and $n_0 = \frac{n+1}{2}$ is the number of leaves.

1. Edge Constraints (E):

$$E_{i,n} = 0 \quad \forall i \quad \text{(Root } n \text{ has no incoming edges)} \tag{9}$$

$$E_{i,j} = 0 \quad \forall i \leq n_0 \quad \text{(Leaves have no outgoing edges)} \tag{10}$$

$$\sum_{j=1}^{n} E_{i,j} = 2 \quad \forall i > n_0 \quad \text{(Internal nodes have 2 children)} \tag{11}$$

$$\sum_{i=n_0+1}^{n} E_{i,j} = 1 \quad \forall j < n \quad \text{(Every node except root has 1 parent)} \tag{12}$$

$$E_{i,j} + E_{j,i} \leq 1 \quad \forall i, j > n_0 \quad \text{(No cycles)} \tag{13}$$

Algorithm: Phylogenetic Tree Structure

2. Ancestor Matrix Constraints (A):

Ensures valid path reachability.

$$A_{i,j} \geq E_{i,j} \tag{14}$$

$$A_{k,j} \geq A_{k,i} + E_{i,j} - 1 \tag{15}$$

$$A_{k,j} \leq A_{k,i} - E_{i,j} + 1 \tag{16}$$

Copy-Number

Root Node & Copy-Number Bounds:

The root clone n represents normal cells (0 for variants, 1 for reference alleles).

$$C_{n,j} = 0 \qquad \forall j \in \{1, \ldots, l+g\} \quad \text{(No variants at root)} \tag{17}$$

$$C_{n,j} = 1 \qquad \forall j \in \{l+g+1, \ldots, l+g+2r\} \quad \text{(Normal copy number)} \tag{18}$$

$$C_{i,j} \leq c_{\max} \qquad \text{(Upper bound for computational tractability)} \tag{19}$$

Breakpoints

Breakpoint Consistency:

Constrains variant copy numbers at breakpoints (b) using segment copy numbers:

$$C_{i,b} \leq \sum_{s=1}^{r} Q_{b,s}\, C_{i,l+g+s} + (1 - D_b)c_{\max} \tag{20}$$

$$C_{i,b} \leq \sum_{s=1}^{r} Q_{b,s}\, C_{i,l+g+r+s} + D_b c_{\max} \tag{21}$$

$$\pi_{i;b} = \frac{F_{i,b}}{\sum_{j=1}^{r} Q_{b,j}\left(F_{i,l+g+j} + F_{i,l+g+r+j}\right)} \tag{22}$$

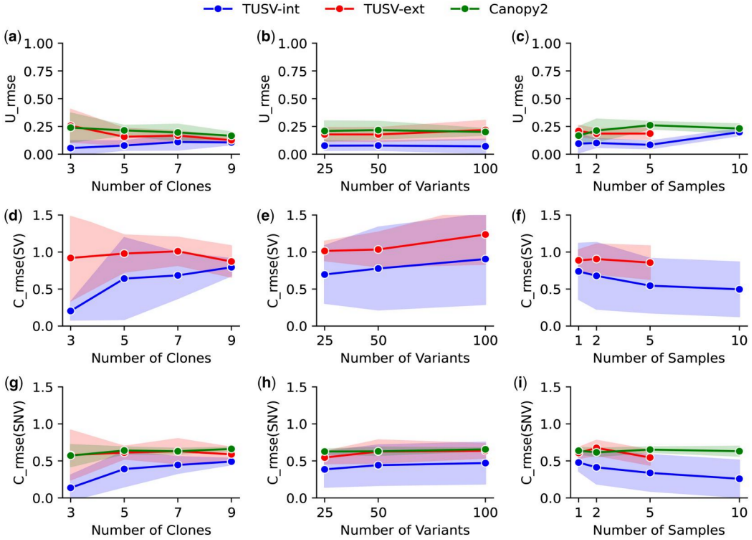
* $\pi_{i;b}$: Mixture ratio ensuring consistency between structural breakpoints and segment copy numbers.

Simulation Results

Key Findings (See Figure 1):

- TUSV-INT significantly outperforms TUSV-ext across most configurations.
- **Clonal Frequencies:** TUSV-INT maintains a strictly lower RMSE (Panels a-c).
- **Structural Variants (SVs):** Canopy2 lacks the ability to predict SVs entirely, whereas TUSV-INT accurately integrates them (Panels d-f).
- Overall performance improves noticeably as the number of available samples (m) increases.

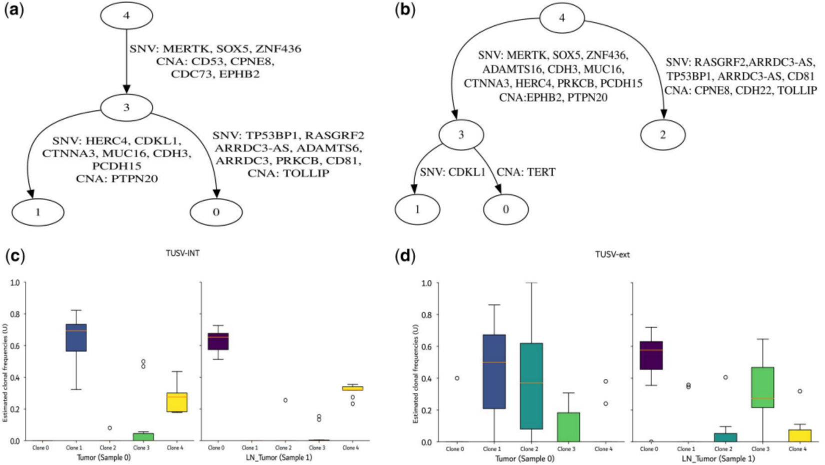
Simulation Results



Case Study: Breast Cancer (BC03)

- **Dataset:** Primary and metastatic samples of patient BC03.
- **Preprocessing:** Variants called using Strelka2, CNVkit, and Numbat.
- **Results:**
  - TUSV-INT inferred **4 distinct clones** (consistent with Canopy2).
  - TUSV-ext inferred 5 clones, highlighting a difference in resolution.
- **Biological Insights:** TUSV-INT suggested a different evolutionary timing for specific mutations (e.g., TP53BP1, TOLLIP), leading to distinct interpretations of concurrent vs. sequential tumor evolution.

Comparing Results



Limitations, Future Work & Conclusion

Current Limitations:

- High computational cost due to NP-hard ILP formulations.
- Lack of comprehensive "gold standard" datasets for direct evaluation.

Future Directions:

- Improving scRNA-seq integration algorithms.
- Extending support for a broader range of variant types.

Conclusion:

- TUSV-INT successfully advances tumor phylogeny by multiomic integration, providing highly reliable clonal deconvolution.

Any Questions?